



OPEN A transparent AI assurance and benchmarking framework for EEG seizure detection on TUSZ seeded with a reproducible gradient-boosting ensemble

Morteza Zabihi^{1✉}, Emily J. Gilmore², Kan Ding³ & Eric S. Rosenthal¹

Reliable evaluation is essential for clinically usable automated electroencephalography (EEG) seizure detection. Although the TUH EEG Seizure Corpus (TUSZ) is the de facto benchmark, studies often use non-canonical splits, mix patients across partitions, or evaluate on seizure-only segments, making results hard to compare. We release an open-source, transparent benchmark for offline binary seizure detection on TUSZ v2.0.3 that strictly follows the official Train/Dev/Eval partition and evaluates in continuous EEG. Each 60-s window (15-s stride) is mapped to expert features and classified by three CatBoost ensembles: a within-window base model and two temporal-context models (full and high-sensitivity). We cast operating-point selection as an alarm-budget decision problem: on the Development set, we jointly choose the score threshold and event post-processing to maximize event sensitivity while enforcing specificity ≥ 0.93 and false alarms ≤ 0.69 per hour, then freeze this decision policy and apply it once to the held-out Eval set. On Eval, we obtain AUROC 0.92 and balanced accuracy 0.83 at the window level and event sensitivity 0.75 at 0.68 false alarms/hour (time-based positive predictive value [PPV] 0.57; recall 0.76). We provide the protocol and results as a reproducible baseline and clinically oriented benchmark for comparing past and future models.

Keywords Seizure detection, Electroencephalogram, TUH EEG Seizure Corpus (TUSZ), Gradient boosting, Temporal context

Electroencephalography (EEG) remains the primary diagnostic modality for seizure diagnosis and monitoring¹. However, EEG monitoring produces large volumes of time series data that require manual review by human experts, who are often unavailable due to geographical, time, or resource constraints. Automated seizure detection systems could improve clinician efficiency and consistency, enable more widespread monitoring, and support large-scale retrospective analyses. However, reliable seizure detection in routine clinical EEG remains challenging due to heterogeneity of seizure patterns across patients, artifacts, broad variation in non-ictal background activity, and extreme imbalance between seizure and non-seizure segments².

The TUH EEG Seizure Corpus (TUSZ)³ has become a broadly utilized open resource for developing and benchmarking seizure detection algorithms. It provides clinically realistic scalp EEG recordings, expert event annotations, and official training/development/evaluation (Train/Dev/Eval) partitions that enforce patient-wise separation. While these design principles should support fair comparisons and realistic estimates of generalizability, the TUSZ is used in a highly heterogeneous manner across the medical literature.

Specifically, many studies focus on seizure subtype classification, often via cross-validation that excludes non-seizure segments^{4,5}. Approaches that address seizure versus non-seizure discrimination often use custom subsets or internal splits that pool the official Train and Eval folders before re-splitting, and include Eval recordings in cross-validation folds⁴. Only a small subset of published work⁶ evaluates seizure detection on TUSZ while (i) respecting the official Train/Dev/Eval partition, (ii) enforcing patient-wise separation, and (iii) reporting both window- and event-level metrics under full class imbalance. As a result, these systems cannot be compared on equal footing or trusted for clinical implementation.

¹Department of Neurology, Massachusetts General Hospital, Harvard Medical School, Boston, MA, USA.

²Department of Neurology, Yale School of Medicine, New Haven, CT, USA. ³University of Texas Southwestern, Dallas, TX, USA. ✉email: mzabihi@mgh.harvard.edu

In this setting, at least as much as new architectures, the field of seizure detection needs a transparent, reproducible benchmark protocol for seizure detection for the TUSZ canon. Such a protocol should (i) make task definition, data partitioning, preprocessing, labelling rules, and evaluation metrics explicit; (ii) operate strictly at the patient level; and (iii) emphasize event-level behavior and clinically meaningful quantities such as false alarms per hour, positive and negative predictive value. A strong, interpretable reference implementation within this framework could then serve as a baseline against which innovative machine-learned classification systems can be fairly judged. In this work, we aim to provide such a benchmark with three open-source artifacts:

1. **A transparent evaluation protocol for binary seizure detection on TUSZ v2.0.3.** We use the official Train/Dev/Eval partition with strict patient-wise separation. We provide comprehensive evaluation metrics, benchmark our results in the context of prior TUSZ-based studies, explicitly distinguishing works that respect the official split from those that use seizure-only subsets or mixed partitions. Under matched, stringent evaluation, our reference system provides a competitive baseline, helping disentangle genuine modeling gains from differences in protocol.
2. **A strong non-deep reference system based on gradient boosting with temporal context.** We construct expert-designed EEG features at the 60-s window level, augment them with temporal-context statistics, and train three CatBoost⁷ ensembles (base, full, and high-sensitivity). Figure 1 shows the proposed pipeline. Beyond interpretability, the non-deep baseline offers flexible decision control and robust performance under extreme imbalance, making it well-suited for use as a reproducible benchmark reference baseline.
3. **Alarm-budget decision framing.** A further source of incomparability is that many works do not fully specify or effectively change the decision policy that converts window scores into seizure events. Yet clinical utility is determined less by threshold-free ranking metrics (e.g., AUROC) than by performance at a fixed alarm burden. We therefore frame seizure detection as an alarm-budget decision problem: on the Dev split, we jointly select the score threshold and event post-processing to maximize event sensitivity subject to explicit constraints on window-level specificity and false alarms per hour. We then freeze this burden-matched policy and evaluate it once on the held-out Eval set, aligning with established concerns about test-set reuse in adaptive data analysis⁸.

Overall, we contribute a transparent, reproducible benchmark for binary seizure detection on TUSZ v2.0.3, including explicit partitioning, labeling, operating-point selection on Dev, and once-only evaluation on Eval. We also release a strong gradient-boosting reference implementation with temporal context and principled event post-processing, intended as a stable baseline for comparing future seizure detection systems under clinically meaningful false-alarm constraints.

Results

Dataset class imbalance

We evaluated on TUSZ v2.0.3³ using the official Train/Dev/Eval partition (Eval is the held-out test set). Across all splits, seizure windows are rare: Train 7.61% (15,509/203,690), Dev 7.17% (7,120/99,252), and Eval 10.21% (2,870/28,110). This strong class imbalance reflects the rarity of seizure relative to non-seizure windows and motivates the use of imbalance-aware training strategies and metrics that go beyond overall accuracy.

Window-level discrimination

Using a Dev-selected operating point ($\tau = 0.44$) obtained by joint search over threshold and event post-processing under specificity/FA constraints, the final three-way ensemble (base/full/high-sensitivity) achieved on Eval: AUROC 0.919 (95% CI 0.877–0.954), AUPRC 0.753 (0.617–0.850), accuracy 0.928 (0.901–0.950), sensitivity 0.701 (0.553–0.822), specificity 0.952 (0.924–0.974), and balanced accuracy 0.827 (0.758–0.880), with 95% CIs. Table 1 reports the bootstrap mean and CI; Fig. 2 reports the full-Eval point estimate of Receiver operating characteristic (ROC, left) and precision-recall (PR, right) curves. Small differences are expected because the bootstrap mean is not identical to the full-sample estimate.

Because seizure windows constitute only 10.21% of the Eval set, the achieved window-level PPV of 0.624 corresponds to an $\sim 6.1\times$ enrichment over the base prevalence (0.624/0.1021). In practical triage terms, this

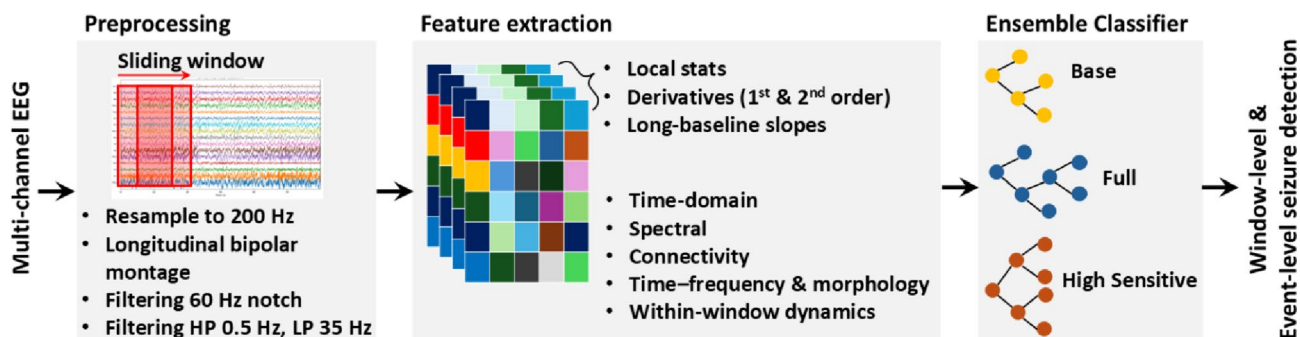


Fig. 1. The proposed seizure detector pipeline.

Metric	Mean	95% CI
PPV (Precision)	0.624	0.503–0.713
NPV	0.967	0.953–0.980
Accuracy (ACC)	0.928	0.901–0.950
Sensitivity (Se)	0.701	0.553–0.822
Specificity (Sp)	0.952	0.924–0.974
Balanced Accuracy (ACCb)	0.827	0.758–0.880
MCC	0.620	0.502–0.702
AUROC	0.919	0.877–0.954
AUPRC	0.753	0.617–0.850
Macro Precision (PrM)	0.795	0.736–0.840
Weighted Precision (PrW)	0.933	0.910–0.952
Macro Recall (RM)	0.827	0.758–0.880
Weighted Recall (RW)	0.928	0.901–0.950
Macro F1 (F1M)	0.809	0.750–0.850
Weighted F1 (F1W)	0.930	0.905–0.950

Table 1. Window-level performance with 95% bootstrap CIs on (Held-out) Eval set with $\tau = 0.44$.

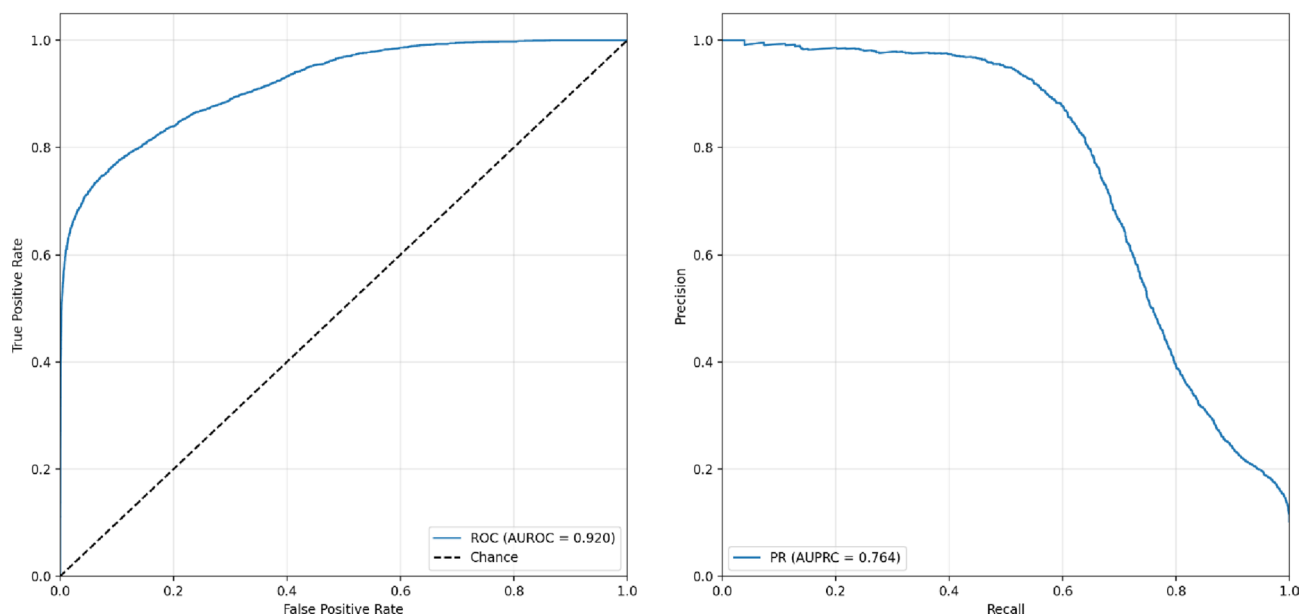


Fig. 2. Receiver operating characteristic (ROC, left) and precision-recall (PR, right) curves on the (held-out) evaluation set using the final 3-ensemble window probability as the score. Curves are computed by pooling all 60-s windows (15-s stride) across Eval recordings and evaluating against window labels; no event-level post-processing is applied in this figure. The dashed diagonal in the ROC plot indicates chance performance.

transforms review from a low-yield search problem into a high-yield screening list, even at conservative specificity.

Patient-wise window-level accuracy

To assess between-patient variability, we computed per-patient window accuracy and balanced accuracy on Eval at $\tau = 0.44$. The model achieved a global (micro-averaged) window accuracy of 0.927 and a global balanced accuracy of 0.827 across 43 patients (836 recordings). Patient-wise accuracy was consistently high but heterogeneous: the median per-patient accuracy was 0.959 (IQR [0.901, 0.989], range [0.687, 1.000], $n = 43$). Because accuracy can be inflated by class imbalance, we also report balanced accuracy; the median per-patient balanced accuracy was 0.863 (IQR [0.616, 0.990], range [0.347, 1.000], $n = 43$).

Event-level detection performance

To assess clinical usability, event-level performance on Eval used the Dev-selected τ and post-processing (smoothing, thresholding, dilation, merging). Results are in Table 2.

Se	FA/H	Time-based PPV	Time-based recall	#Files	#Seizure events	Detected events	FP	Hours
0.749	0.682	0.568	0.761	836	291	218	87	127.58

Table 2. Event-level performance on the TUSZ Eval (held-out test) set. Se: Sensitivity; FA/H: False alarms per hour; PPV: Positive predictive value; FP: False positive.

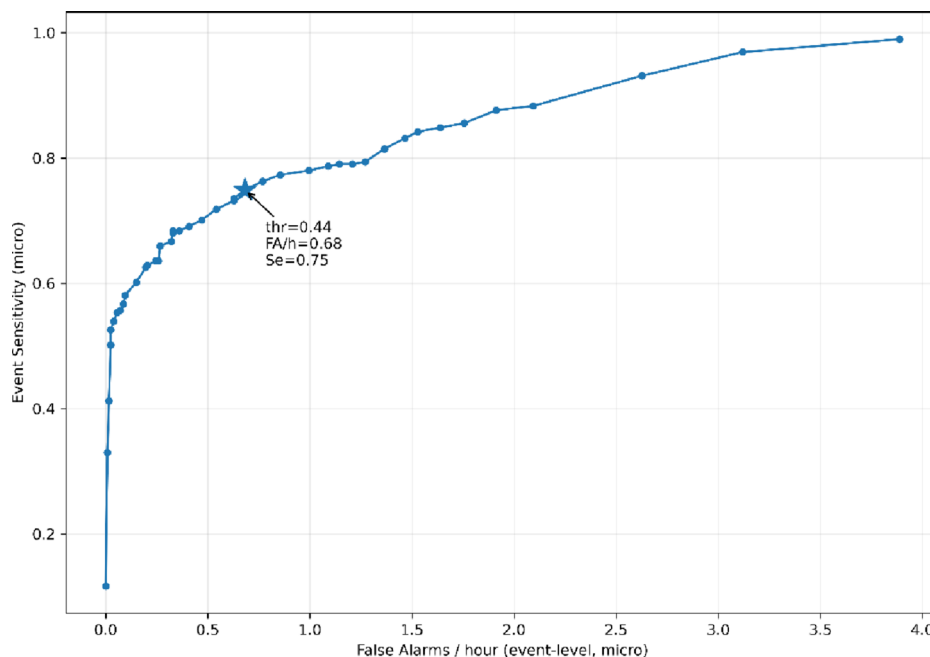


Fig. 3. Beyond the window-level ROC curve, an event-level Free Receiver Operating Characteristic curve (FROC) is a critical benchmarking component for detailing event-level performance trade-off between event sensitivity and false alarm rate per hour as the decision threshold is varied. This summarizes the end-to-end alarm burden after post-processing; window-level ROC/PR cannot be ‘pro-rated’ to FA/h because sliding-window overlap and event merging change how FP windows translate into alarm events. Each marker details a different operating threshold from the fixed post-processing pipeline. The star highlights a chosen optimal operating point at a seizure probability threshold of 0.44, which provides a Sensitivity of 0.75 at 0.68 false alarms per hour. This threshold is chosen for an offline electrodiagnostic screening setting (i.e., EEG reviewer), but an alarm for the clinical setting (i.e., nurse or provider) would likely be chosen at a different threshold.

The detector obtained event sensitivity 0.749 (218/291 seizures detected), with 0.682 false alarms per hour (87 false-positive events over 127.58 h). The time-based positive predictive value is 0.568 (56.8% of predicted seizure time overlaps true seizure time), and time-based recall is 0.761 (76.1% of total seizure duration is covered by predicted intervals). Time-based PPV and recall are duration-weighted overlap metrics: they quantify the fraction of predicted seizure *time* that overlaps true seizure time (PPV) and the fraction of true seizure *duration* covered by predictions (recall), complementing event-count sensitivity.

To summarize how well the proposed model detects seizure events while controlling the rate of false alarm events per hour of EEG, we plot an event-level Free Receiver Operating Characteristic curve (FROC) (Fig. 3). While Fig. 2 shows how well the model ranks seizure vs. non-seizure windows, Fig. 3 shows how that ranking becomes clinically meaningful alarm events per hour after post-processing, which cannot be inferred from window-level ROC/PR because overlapping windows and event merging fundamentally change the relationship between false-positive windows and false alarm events. Moving along the curve corresponds to changing the operating threshold: lowering the threshold increases sensitivity (more events detected) but also increases FA/h, whereas raising it reduces false alarms at the cost of missed events. The annotated point at $\tau = 0.44$ indicates our selected trade-off (0.68 false alarms/hour while detecting 75% of seizure events). The “knee” region around this point suggests achieving higher sensitivity requires a disproportionately large increase in false alarms, supporting our choice of operating point.

Per-patient and per-seizure heterogeneity and tail risk

Performance was heterogeneous on the TUSZ held-out Eval set. Among 291 seizure events on the Eval set, median per-patient event sensitivity was 0.956 (IQR 0.446–1.000; range 0–1). False alarms were typically low at the recording level (median 0.0 FA/h; range 0–20), but a small number of recordings had high false-alarm

burden (up to 20 FA/h), indicating a clinically important long-tail failure mode. Also, per-patient false-alarm burden showed a long tail across all 43 Eval patients (median 0.161 FA/h; IQR 0–1.031; range 0–6.716).

Sensitivity decreased with higher seizure burden (median sensitivity: 1–2 seizures/patient 1.000; 3–5 seizures/patient 1.000; ≥ 6 seizures/patient 0.769). Shorter seizures were harder: 30–90 s events had sensitivity 0.333 (4/12) vs. 0.767 (214/279) for > 90 s; no < 30 s events remained after merging.

Temporal precision (post-processed detector behavior)

Detection latency was defined as the time from annotated seizure onset to the start of the first predicted interval that overlapped the seizure (any overlap); predicted intervals that began before onset were assigned a latency of 0 s. Among detected seizures on the held-out TUSZ Eval set ($n = 218$), median latency was 0.0 s (IQR 0.0–0.0; range 0.0–1200.0). Notably, the near-zero latency should not be interpreted as preictal prediction. Because the event construction uses centered smoothing and temporal dilation/merging, predicted intervals can extend earlier than the true onset, producing apparent ‘early’ detections that reflect conservative interval expansion rather than onset anticipation. We therefore treat latency here as a temporal-coverage property of the post-processed detector in an offline setting.

Early detections were common: 169/218 (77.5%) had a predicted interval already active before onset, whereas 49/218 (22.5%) began after onset. Stratified by duration among detected events, median latency was 0.0 s (IQR 0.0–0.0) for both 30–90 s seizures ($n = 4$ detected) and > 90 s seizures ($n = 214$ detected), with rare extreme late outliers (max 1200.0 s).

What drives predictions: feature importance + SHAP

Across ensemble models, the most influential predictors reflected spectral organization and its short-term instability within each recording. The classifier relied strongly on the dominant low-frequency rhythm (1–15 Hz) estimated from the Welch power spectral density (regional median dominant-peak frequency), and on theta-band power expressed as an upper-tail statistic (95th percentile) and its local variability, indicating sensitivity to transient increases in theta activity and fluctuations around a local baseline. Additional high-importance contributors captured alpha “burstiness”, quantified as the fraction of time the alpha-band amplitude envelope (Hilbert envelope of 8–13 Hz bandpass) exceeded a robust per-channel threshold, and (importantly in the temporally enriched feature set) the short-term volatility of this burstiness via derivative- and context-window summaries. The model also emphasized signal complexity/morphology proxies, including the count of prominent peaks in the 5-s subwindow line-length trajectory (a measure of waveform roughness/absolute derivative activity) and its local mean, along with alpha-band spectral shape (spectral flatness/crest), consistent with decisions being driven by event-associated shifts in rhythmicity, spectral concentration, and time-localized changes in these properties.

In practical terms, the model is looking for windows where the EEG’s “main rhythm” shifts and where that rhythm becomes less steady across nearby time windows. When the dominant frequency changes and theta activity shows brief surges (especially strong bursts rather than a uniform increase), the model treats that as a meaningful signature. It also pays attention to whether the alpha rhythm comes in irregular bursts and whether those bursts start fluctuating more from one short segment to the next. Finally, features based on line length and peak counts indicate that the model is sensitive to periods where the signal becomes “sharper” or includes more rapid changes, which often accompany abnormal events compared with more regular background activity. Figures 4 and 5 present the aggregated feature importance by frequency band and feature family for the 30 most influential features.

To further interpret the trained CatBoost base classifier (no temporal context features), we computed SHAP (SHapley Additive exPlanations)⁹ values on the held-out window eval set. SHAP provides an additive decomposition of each prediction into per-feature contributions, enabling both global ranking (mean |SHAP|) and directionality (whether high or low values increase the model score). The resulting global beeswarm plot (Fig. 6) shows that the model primarily relies on rhythmic content, especially dominant frequency and theta/alpha-band structure, and on how stable or variable these signals are within the 60-s window (line-length variability/peak structure and theta “volatility” features). This is aligned with CatBoost feature importance. Directionally, windows with higher dominant frequency tend to push predictions toward the positive class, while theta synchrony/asymmetry features and alpha spectral shape modulate the decision in a way consistent with shifts in rhythmic organization and inter-hemispheric coupling, suggesting the model is detecting event-related changes in oscillatory dynamics rather than relying on gross amplitude alone.

Risk profile (qualitative analysis)

In visual inspection of representative errors, false positives were often associated with prominent, synchronous, sharp transients of large amplitude appearing across many channels, consistent with non-cerebral artifact (e.g., movement/electrode-related activity) that can resemble ictal morphology. In contrast, false negatives frequently contained slower, higher-amplitude rhythmic activity confined to a subset of channels, suggestive of a more focal pattern (or focal artifact) that may be underemphasized by the handcrafted feature set and the model’s channel-aggregation strategy (Fig. 7). Further misclassified examples are provided in the *Supplementary Materials*.

Discussion

The scientific community has raised a call to action to develop a nationwide network of health AI assurance laboratories¹⁰, of which key components prior to any deployment include rigorous model evaluation on established cohorts, transparent model performance reporting, and analysis of potential benefit. Towards this goal for seizure detection, we present a rigorous, transparent AI assurance framework with comprehensive benchmarks reported in an exemplar open-source model card for the flagship Temple University Hospital

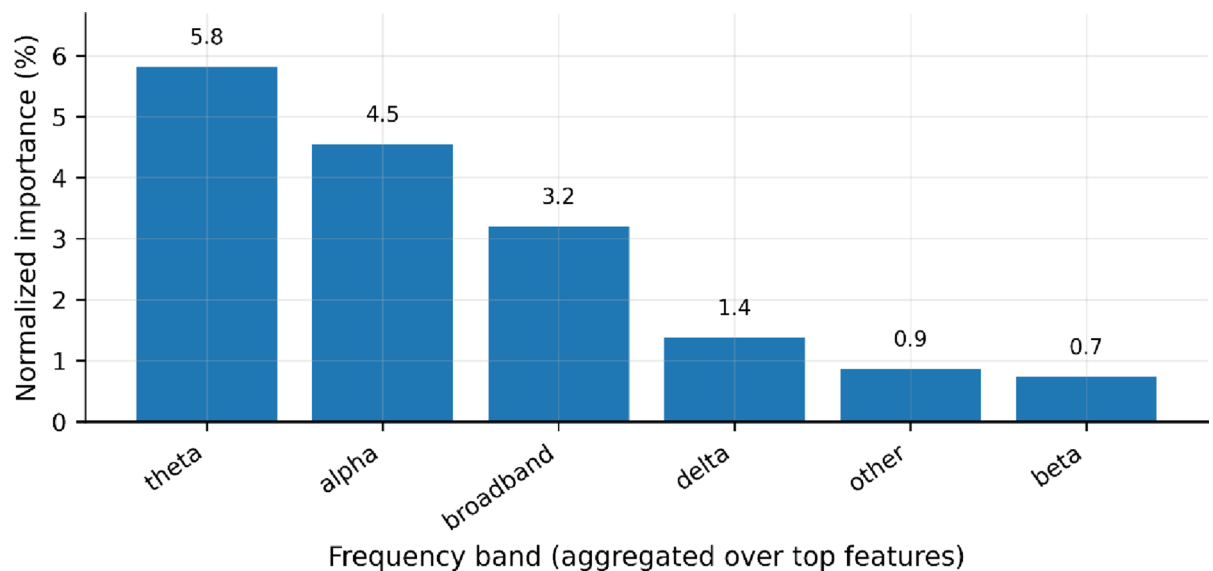


Fig. 4. Aggregated feature importance by frequency band for the 30 most influential features. Importances are CatBoost *PredictionValuesChange* values computed from the trained ensemble and summed within band categories (delta, theta, alpha, beta, broadband, and other). Values are normalized so that total importance across all features equals 100%, highlighting that theta/alpha-band and broadband dominant-frequency features contribute most to classification.

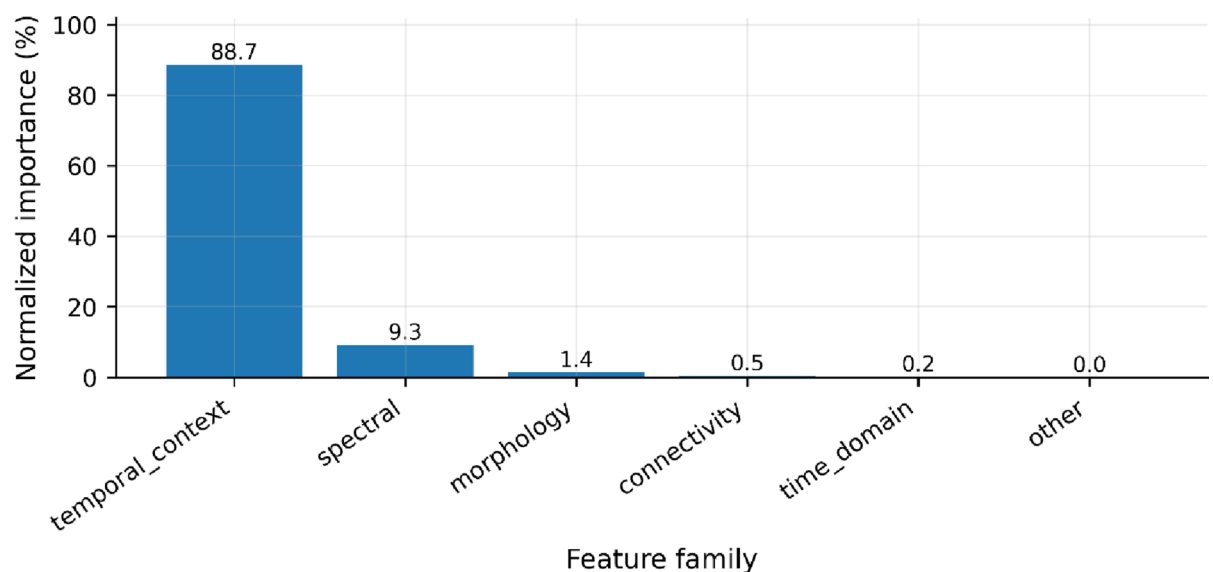


Fig. 5. Aggregated feature importance by feature family. CatBoost *PredictionValuesChange* importances are summed across all features within families (temporal-context, spectral, morphology, connectivity, and time-domain) and normalized to 100% total importance.

EEG Seizure Corpus. We developed and tested a gradient boosting ensemble model with temporal context, stress-testing it under this comprehensive set of performance metrics relevant to model type and use case, and contextual interpretability analyses. In this setting, we trained, developed, and evaluated a gradient boosting ensemble for seizure detection on the TUH EEG Seizure Corpus, employing a set of controls to limit overfitting while ensuring fair evaluation. The system combines expert features, temporal context, and multiple CatBoost ensembles, with a joint search on the development set to select both the decision threshold and event post-processing parameters under explicit specificity and false-alarm constraints.

Performance-burden trade-offs, salient failure modes, and intended-use alignment

The resulting system exhibits several desirable properties while transparently demonstrating opportunities for improvement under the intended stress test of the benchmark. In terms of classification performance under

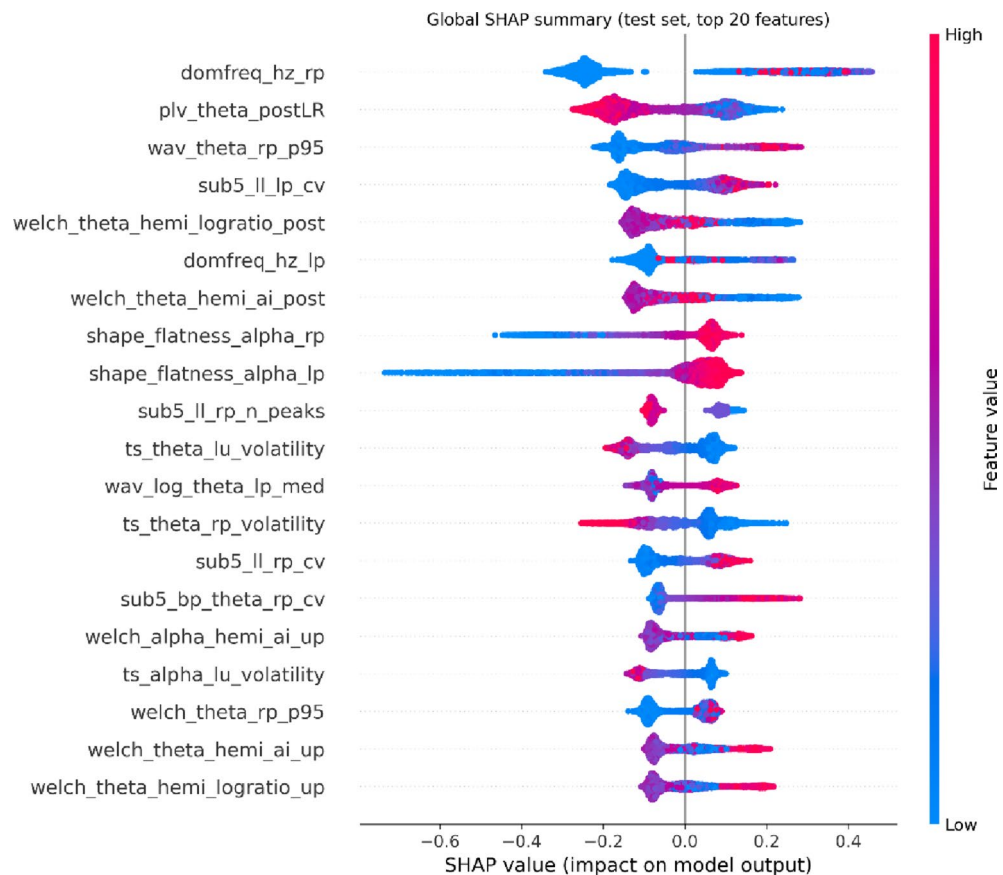


Fig. 6. Global SHAP summary on the held-out window set (top 20 features, CatBoost base ensemble). Each point represents one 60-s EEG window; the x-axis shows the SHAP value (feature contribution to the model output), where positive values push the prediction toward the positive class and negative values push it toward the negative class. Point color encodes the feature value (blue = low, pink = high). Features are ranked by mean absolute SHAP value across all windows. The highest-impact predictors describe: (i) the dominant oscillatory frequency in the right posterior region (dominant peak within 1–15 Hz), (ii) theta-band phase synchrony between left and right posterior channels (posterior L-R theta PLV), (iii) high-end (p95) theta power/energy in the right posterior region (wavelet/Welch summaries), (iv) temporal irregularity of waveform complexity within the window (e.g., coefficient of variation and peak counts of 5-s subwindow line length), (v) theta hemispheric asymmetry in posterior regions (log-ratio and asymmetry index), and (vi) alpha-band spectral shape in posterior regions (spectral flatness; lower flatness indicates a more “peaked”/rhythmic spectrum). Together, the plot indicates that rhythmic structure, cross-hemispheric theta organization, and within-window nonstationarity account for most of the model’s global decisions. The list of abbreviations of the top 20 features is provided in the *Supplementary Materials*.

realistic class imbalance, the model achieved an AUROC of 0.92 and a balanced accuracy of 0.83 on the Eval (held-out) set, indicating that the model discriminates seizure from non-seizure windows reliably despite the rarity of seizures. However, discrimination does not fully translate to error-free operation at a fixed threshold: at the selected operating point, the window-level sensitivity is 0.70, implying that roughly 30% of seizure windows are missed, and the precision is 0.62, indicating a meaningful fraction of false-positive seizure windows under strong class imbalance. These trade-offs motivate reporting operating-point metrics (Se/Sp/PPV) alongside threshold-free AUROC/AUPRC, and reinforce that intended use (e.g., triage vs. autonomous alarming) must dictate where on the ROC/PR surface the system should operate.

In terms of the benchmark’s requirement for both minimum specificity and an upper bound on false alarms during operating-point selection, the detector operates in a relatively conservative regime for offline triage, such that the trained model achieved an approximate specificity of 0.95 and a false alarm rate of 0.68 per hour. Under these constraints, the detector yields a lower alarm burden than the range reported in most of the prior works^{4,5,11,12}, if reported, and may be acceptable for *offline triage* workflows (e.g., prioritizing segments for review). Nevertheless, the stress test of the benchmarking leaves clear room for future improvement, as well as room for interpretation of what the acceptable false-alarm rates should be for different clinical applications, such as prioritizing a reading list for high-risk studies versus autonomous interpretation in an emergency department, far-forward military theatre, or rural environment lacking immediate availability of expert clinical neurophysiologists. Additionally, we demonstrate the value of examining the proportion of seizures detected;

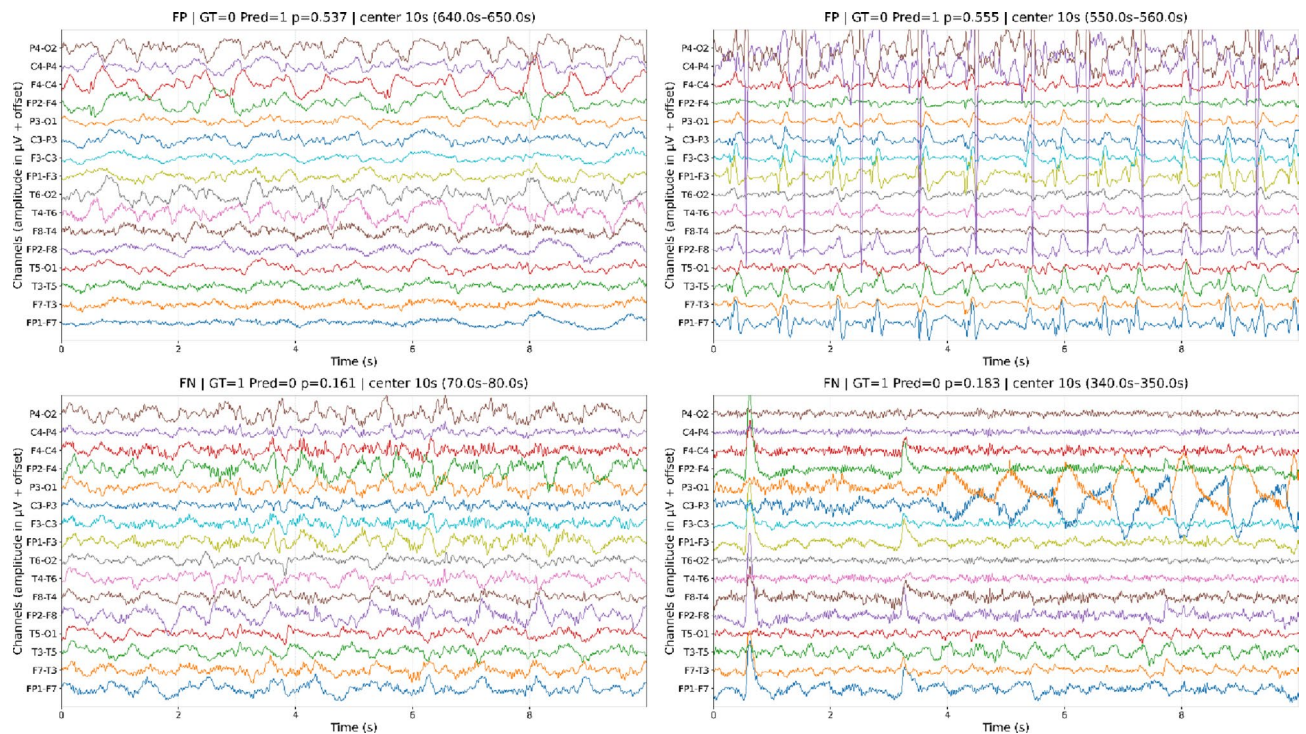


Fig. 7. Qualitative window-level misclassification evaluation using raw EEG. For each misclassified 60-s analysis window, we plot only the 10-s center segment (25–35 s within the window) to match conventional EEG readability and avoid excessive temporal compression. Traces show the multi-channel EEG (scaled to μV with vertical offsets) for false positives (FP; predicted seizure, ground truth non-seizure) and false negatives (FN; missed seizure). Each panel header reports the error type, ground-truth label (GT), predicted label (Pred), model probability (p), and the time range of the displayed 10-s segment. Additional examples are depicted in *Supplementary Fig. S2 and S3*.

documenting a sensitivity of 0.75 and a time-based recall of 0.76, we show that a system may recover a majority of seizures and seizure duration while keeping false alarms at a moderate rate.

However, the event-level results also reveal clinically important failure modes: performance is heterogeneous across patients and recordings, with a small subset contributing disproportionately to false alarms and with markedly reduced sensitivity for shorter seizures (e.g., 30–90 s events). Moreover, because event construction includes temporal smoothing and dilation, many “early” detections reflect conservative interval expansion rather than true preictal prediction; this improves coverage and sensitivity but can reduce temporal precision and can inflate the fraction of detections that appear at or before onset.

The benchmarking underscores model design choices around temporal context, ensembling, and operating-point selection for non-deep detectors. We include a temporally enriched feature set that augments expert window features with local context and derivative summaries to capture short-term nonstationarity across adjacent windows. In addition, selecting the window threshold jointly with event post-processing on the development set under explicit specificity and false-alarm constraints yields more clinically meaningful event-level behavior than naive thresholding. Finally, averaging complementary ensembles (base, full, and a sensitivity-optimized variant) helps balance window-level discrimination with event-level sensitivity and false-alarm burden at the chosen operating point.

The significance of this evaluative framework is highlighting the ease of deploying and inspecting a model utilizing gradient boosting on expert-based features. For example, key drivers of the present model were dominant frequency changes, theta/alpha-band activity, morphology, and connectivity, particularly in posterior regions (in this dataset and montage), which enable evaluation of face validity regarding features with expected importance in diagnosing focal and generalized seizures.

The heterogeneity analysis shows that, even in the setting of a high median per-patient event sensitivity (~ 0.90), a non-trivial subset of patients may experience substantially lower sensitivity and/or a higher false-alarm burden, indicating that deployment would require careful monitoring of per-patient performance and potentially adaptive tuning. Specifically, seizures shorter than 90s are detected much less frequently than longer events, underscoring that short focal seizures as an important failure mode requiring awareness for any clinical deployment and benefiting from future advances. These findings, taken together with the latency results, reinforce how a detector may be best suited for retrospective triage and workload reduction rather than autonomous deployment. In the present model under evaluation, most detected seizures are flagged at or near onset, but the number of missed or delayed events currently limits its use as a robust early-warning system.

Comparison with prior work on TUSZ

When interpreting our results, it is important to account for the substantial variation in how the TUSZ corpus has been used across the literature in terms of task definition, data partitioning, and evaluation granularity.

Among studies that are most comparable to ours, Wong et al. (2025) used a CNN-Transformer architecture on TUSZ v2.0.1 using the official Train + Dev partitions and evaluated on the held-out Eval set at both segment- and event-level⁶. At the segment-level, this approach yielded a reported AUC of 0.93, accuracy of 0.88, sensitivity of 0.82, and specificity of 0.88. However, event-level sensitivity reached 0.97 at about 1.7 FA/h. While Wong et al.'s approach was tuned for rapid event alarming, our 60-s feature-based ensemble emphasized lower false-alarm burden and strong window discrimination (AUROC 0.919, Sp=0.952), achieving event-level Se=0.749 at 0.682 FA/h.

Liu et al. (2025) evaluated seizure vs. non-seizure discrimination on 4-s segments (not events) and reported 92.6% sensitivity, 92.5% specificity, 92.5% accuracy, and F1=0.76 using a meta-learning-guided decision-tree ensemble¹³. Because their procedure did not follow the canonical TUSZ Train/Dev usage, direct numerical comparison to detectors tuned under the official split should be interpreted cautiously. This mismatch illustrated why a standard evaluation framework is necessary, because convenient custom splits can result in optimistic estimates via leakage or overfitting.

Several other works reported very high accuracy on TUSZ, but under substantially different and less stringent protocols. Khosla et al. (2021)⁴ and Kashefi Amiri et al. (2025)⁵ both perform multi-class seizure-type classification on seizure-only segments rather than seizure detection in continuous, highly imbalanced EEG. Khosla et al.⁴ pool the official Train and Eval sets, extract short seizure-only segments, and perform 5-fold cross-validation at the segment level, obtaining near-100% accuracy and F1 for six seizure types. Because Train/Eval are patient-disjoint in TUSZ, pooling removes the held-out patient split and can yield optimistic results, especially when multiple segments come from the same seizure.

Kashefi Amiri et al.⁵ used only subjects from the TUSZ dev folder, applying 10-fold cross-validation on 1-s windows. They reported 94% accuracy and 85% F1 across eight seizure types with a 1D CNN-LSTM model. However, this evaluation was a segment-level cross-validation with random shuffling and did not clearly enforce record-wise or subject-wise separation (nor the standard TUSZ Train/Dev/Test protocol). In EEG, random cross-validation over short windows can place highly correlated segments from the same recording/patient into both train and test folds, a data leakage which may yield optimistic estimates relative to strict held-out subject/record evaluation.

Statsenko et al. (2023) focused on a hand-picked subset of 33 TUSZ patients with long seizure durations, performing window-based 5-fold cross-validation on 1-s STFT inputs¹¹. They obtained 88% sensitivity and 91% specificity for binary seizure detection and 96% accuracy for eight-way seizure-type classification, again on a seizure-rich subset rather than the full corpus. Similarly, Sánchez-Hernández et al. (2024)¹² evaluated a patient-specific seizure vs. non-seizure CNN on a preprocessed subset of TUSZ (21 patients). They reported on TUSZ averaged across patients, best performance of 0.69 sensitivity, 0.58 specificity, 0.64 accuracy, and 0.62 F1 (AUC 0.74).

These configurations (*Supplementary Materials*, Table S2), like those above, provide useful information about model capability but differ from the training, development, and evaluation framework we advocate in at least one of the following: (i) using seizure-only or heavily enriched subsets; (ii) pooling official train/test splits; (iii) evaluating with cross-validation across windows instead of a fixed, patient-disjoint Eval set; (iv) reporting segment-level rather than event-level metrics; and (v) reporting limited performance measures instead of providing a comprehensive set of metrics to shed light on different aspects of performance and enable a fair evaluation. In this context, our gradient boosting ensemble achieved discrimination performance comparable to state-of-the-art deep learning methods that respect the official split (e.g., Wong et al.⁶), while deliberately operating in a high-specificity, low-false-alarm regime on the official Eval recordings. At the same time, the gap between our results and those obtained under more permissive protocols highlights how choices in task definition and evaluation strategy can potentially influence headline numbers on TUSZ.

Across the TUSZ literature, headline event sensitivities and false-alarm rates depend not only on the classifier but also on the downstream decision policy: score calibration (if any), threshold selection split (held-out Dev vs. reusing test data), and event post-processing details (smoothing, hysteresis/dilation, merge gaps, minimum duration, and the overlap rule used to count detections). These components are often under-specified or not standardized across studies, making it difficult to fully 'normalize' reported FA/h and sensitivity under a single common alarm-budget policy. We therefore restrict our direct comparisons to studies that clearly evaluate on the official held-out Eval set and emphasize our contribution as providing an explicit Dev-selected, frozen policy that future work can adopt and report in full for fair burden-matched benchmarking.

Additionally, prior work on TUSZ spans a range of input representations, including raw EEG time-series, time-frequency maps (e.g., STFT/spectrograms), and wavelet-based representations paired with deep architectures, as well as smaller sets of handcrafted features in classical ML pipelines. In contrast, our reference system uses an explicit, clinician-aligned feature set that summarizes spectral organization (band powers, dominant frequency, spectral shape), morphology/complexity proxies (line length and its within-window dynamics), and connectivity statistics (e.g., phase-locking/coherence) aggregated into coarse anatomical regions, augmented with offline temporal-context summaries. This design encodes EEG waveforms into an interpretable and spatially finer representation and provides a transparent and controllable operating behavior under extreme imbalance, while also potentially under-representing fast ictal patterns that may be better captured by higher-bandwidth features.

Complementary community efforts aim to standardize seizure-detector validation beyond any single dataset. SzCORE (Seizure Community Open-Source Research Evaluation) proposes a unified framework with recommendations for datasets and standardized data/annotation formats (EEG-BIDS and HED-SCORE-compatible annotations), cross-validation practices to preserve subject independence, and consistent reporting

of sample-based and event-based performance metrics. It also introduces an ‘EEG 10–20 seizure detection benchmark’ built from multiple public datasets converted into a standardized format¹⁴, and provides open-source tooling and an online platform to facilitate reproducible scoring and community comparison. Our contribution is complementary and narrower in scope: we anchor a transparent, patient-disjoint Train/Dev/Eval workflow specifically on TUSZ v2.0.3 and provide a reproducible baseline plus a Dev-selected, frozen alarm-budget decision policy.

Contextualizing FA/h targets with commercial detectors

Our chosen operating point ($Se = 0.75$ at 0.68 FA/h) yields an alarm burden comparable in scale to commercially deployed seizure-detection software evaluated in EMU video-EEG. In Koren et al. (2021), mean false-alarm rates for commercial tools were on the order of ~ 0.2 – 0.9 /h, with detection rates roughly ~ 68 – 82% depending on the package and configuration¹⁵. More recently, a large head-to-head EMU benchmark (Koren et al., 2025) showed a pronounced sensitivity-false-alarm trade-off: Persyst 14 achieved very low mean FPR (~ 0.07 – 0.17 /h) at lower mean detection rates (~ 67 – 71% per patient), whereas encephalis 2.0 achieved higher mean detection rates (~ 81 – 89% per patient) at higher mean FPR (~ 0.5 – 0.93 /h)^{14,16}. In continuous inpatient EEG, Ganguly et al. reported that Persyst had seizure-level sensitivity 0.50 with seizure-level PPV 0.60 in a pilot cohort, while record-level sensitivity was higher (~ 0.78 – 0.91) but PPV was substantially lower (0.40 in the pilot cohort and 0.08 in the expanded cohort)¹⁷. Direct numerical comparison remains limited because these studies used different cohorts (single-center EMU/ICU populations, often seizure-enriched) and different detection definitions (e.g., seizure-epoch matching vs. overlap-based event scoring), whereas we benchmark on TUSZ continuous EEG under the official patient-wise Train/Dev/Eval split. Nevertheless, these commercial evaluations support the need to select operating points under an explicit false-alarm budget and to report long-tail false-alarm behavior as a key deployment risk.

Limitations and next steps

The current system relies on a fixed preprocessing pipeline and a particular feature set; alternative montages and artifact handling strategies could further improve performance. The three ensembles are trained independently and averaged with equal weights; learning adaptive per-patient or per-record weights, or introducing a calibrated meta-learner on out-of-fold predictions, could better exploit complementary strengths. Our experiments are confined to TUSZ; assessing generalization to other EEG datasets and developing domain adaptation strategies remain important steps toward deployment across institutions. We did not stratify performance by seizure type, so potential differences in detection performance across focal and generalized seizures or across anatomical lobes remain to be characterized. We also did not perform a formal sensitivity analysis of the window-labelling rule (the choice of overlap thresholds for assigning seizure vs. non-seizure labels to 60-s windows). Because successive windows overlap by 75% , small changes in these thresholds primarily affect a limited number of boundary windows and are therefore unlikely to drastically alter overall performance; however, they may complicate direct comparisons with methods that use different window lengths, strides, or labelling conventions.

Moreover, the observed long-tail false-alarm behavior is clinically important because a globally acceptable mean FA/h can still hide a small subset of recordings/patients with unacceptable alarm burden. Without additional experiments on multicenter cohorts, we cannot conclusively determine which patients are ‘predictable’ high-FA outliers and what the plausible contributors are, such as artifact, channel dropouts, montage irregularities, and periodic or sharply contoured non-ictal patterns that resemble ictal morphology in the handcrafted feature space. Importantly, long-tail FA/h is jointly determined by the window-score distribution and by post-processing/threshold choices: operating points selected to satisfy a global FA/h constraint may still yield disproportionate alarms in a minority of cases. Practical mitigation strategies (future work within this benchmark) include artifact-aware gating, abstention, calibration using an initial presumed non-seizure segment, and controlled per-patient threshold adaptation evaluated under a strict protocol to avoid leakage.

Despite these limitations, the present work demonstrates that, under realistic and fully transparent use of TUSZ, a gradient boosting ensemble can deliver strong seizure detection performance. Beyond the specific model family, the primary contribution of this work is a burden-matched evaluation paradigm. We show that event-level performance is a property of the *end-to-end decision system*, i.e., model scores plus post-processing, not of the classifier alone, and that fair comparison requires fixing an operating policy under shared constraints. This “alarm-budget” framing shifts the benchmark from architecture-centric claims to clinically grounded decision-making: *how much seizure coverage a method can deliver for a given, explicitly bounded alarm burden*.

Methods

The task is defined as offline binary seizure vs. non-seizure classification on 60-s windows with 15-s stride. Models were trained on Train; the operating threshold and event post-processing were selected on Dev; final metrics were reported once on Eval. The model outputs a seizure probability per 60-s window; event post-processing converts window scores into predicted seizure intervals.

Dataset

We used the TUH EEG Seizure Corpus (TUSZ) v2.0.3³, a large clinical EEG dataset with expert-annotated seizures and an official partitioning into Train, Dev, and Eval sets. Each recording comprises multi-channel scalp EEG sampled at standard clinical rates and accompanied by channel montages and seizure annotations with onset and offset times. For this study, we used the Train split for model fitting (with internal patient-wise cross-validation), the Dev (development) split for threshold and parameter selection, and the Eval split as a held-out test set for final evaluation.

Preprocessing

The preprocessing pipeline standardizes sampling rate, montage, units, and frequency content before windowing and labelling. Each recording is resampled to a uniform sampling frequency of 200 Hz, ensuring consistent temporal resolution. To obtain a standardized spatial representation, we transformed the original referenced channels into a fixed-order longitudinal bipolar montage. We defined 18 bipolar derivations: Left temporal chain: FP1-F7, F7-T3, T3-T5, T5-O1; Right temporal chain: FP2-F8, F8-T4, T4-T6, T6-O2; Left parasagittal chain: FP1-F3, F3-C3, C3-P3, P3-O1; and Right parasagittal chain: FP2-F4, F4-C4, C4-P4, P4-O2. This yields a matrix of shape $n_pairs \times n_times$ with a fixed channel ordering. The transformed bipolar data were stored as 32-bit floating point values in volts and used for all subsequent filtering and windowing.

To reduce line noise and restrict the analysis to conventional EEG frequencies for seizure detection, we applied a sequence of zero-phase digital filters to the bipolar signals, each applied with forward-backward filtering. First, we applied an IIR notch filter centered at 60 Hz to attenuate mains interference. Second, we removed low-frequency drifts with a first-order Butterworth high-pass filter at 0.5 Hz. Finally, we attenuated high-frequency activity and residual muscle artifact with a fourth-order Butterworth low-pass filter at 35 Hz. We chose a 35 Hz cutoff as a compromise between retaining most conventional ictal EEG activity in the delta-beta range and suppressing high-frequency muscle artifacts; very fast ictal patterns may therefore be attenuated to some extent.

Windowing and seizure-label assignment (continuous EEG)

We interpret any annotation whose label either (i) ends with the suffix “sz”, e.g., fnsz, gnsz, or (ii) equals “seiz” or “seizure” as a seizure event. All other labels are treated as non-seizure. After filtering, each recording is segmented into 60-s windows with 75% overlap, corresponding to a 15-s stride. For each window w , with start time t_s and end time $t_e = t_s + 60$ seconds, we computed its temporal overlap with each merged seizure interval $[s_i, e_i]$. The total overlap of the window with interval i is

$$\text{ovl}_i = \max(0, \min(t_e, e_i) - \max(t_s, s_i)).$$

We then applied a combined, relative-overlap labelling rule designed to avoid labelling windows as seizures when they contain only very brief seizure fragments:

- For each seizure interval $[s_i, e_i]$, we computed its duration $L_i = e_i - s_i$.
- For that interval, we defined a per-interval threshold.

$$\tau_i = \min(\theta_{\text{win}} \cdot 60\text{-s}, \theta_{\text{seiz}} \cdot L_i),$$

where θ_{win} is a fraction of the 60-s window and θ_{seiz} is a fraction of the seizure duration. In all experiments, we used $\theta_{\text{win}} = 0.15$ (9s) and $\theta_{\text{seiz}} = 0.20$ (20% of the seizure’s length). Thus, a window was assigned label 1 (seizure) if there exists at least one seizure interval i for which $\text{ovl}_i \geq \tau_i$; otherwise, it was assigned label 0 (non-seizure).

This rule ensures that: (i) Windows with no overlap with any seizure are labelled non-seizure. (ii) Windows that touch a seizure only briefly (e.g., at onset/offset) and cover <9 s of that seizure are labelled non-seizure. (iii) For short seizures, the threshold scales with the seizure duration, requiring coverage of at least 20% of that interval to be considered a seizure.

Expert feature set

We computed a comprehensive set of hand-crafted features per 60-s window, summarizing amplitude, spectral content, connectivity, and temporal dynamics across four anatomical regions (left/right anterior, left/right posterior). Time-domain features included root-mean-square amplitude, line length, zero-crossing rate, spike-wave indices, and Hjorth parameters. Spectral features encompassed band-limited powers (delta, theta, alpha, beta), band ratios, spectral entropy, 1/f-like slopes, and alpha-peak frequency. We further derived connectivity measures (phase-locking values and coherence in canonical bands), wavelet-based multiscale energies and entropies, and envelope-based descriptors of oscillatory amplitude and burstiness. Finally, within-window temporal-evolution features summarized how bandpower and simple time-domain measures changed over shorter (5 s) segments within each 60-s window. Full feature definitions and parameter settings are provided in the *Supplementary Material*.

All of the above quantities are concatenated to form a single feature vector. This “base” feature vector serves directly as input to the base gradient boosting ensemble and also as the foundation for the temporal context features described in the following section.

Temporal context features

In addition to per-window base features, we construct temporal context features designed to capture the evolution of EEG statistics around each window. For each numeric base feature f and for each file (recording), we define a symmetric temporal neighborhood of size $\pm k$ windows around each target window (here $k = 7$), corresponding to a temporal span that depends on the window stride. For windows within k positions of the start or end of a recording, the temporal neighborhood was truncated to the available windows; we did not pad across recording boundaries or with artificial values. All temporal context statistics were therefore computed using the variable-size neighborhood actually present in the recording. Because the temporal neighborhood includes future windows, these temporal-context features are intended for offline analysis rather than real-time deployment.

For each window and base feature f , we compute: (i) Local window statistics over the neighborhood $[i - k, \dots, i + k]$: Mean, minimum, maximum, standard deviation, range (max-min), z-score of f_i relative to the local mean and standard deviation. (ii) First-order temporal derivatives: Approximate derivative df/dt using the difference between consecutive windows normalized by the time difference between their start times, then local mean and standard deviation of this derivative over the same neighborhood. (iii) Second-order derivatives: Approximate second derivative via a central difference scheme, normalized by the squared median stride within the recording. (iv) Long-baseline slopes: Slope of f between windows $i - k$ and $i + k$, normalized by the time interval between their start times. All these derived quantities are computed per file to avoid any information leakage across recordings or patients.

CatBoost committees and three-ensemble fusion

We trained three patient-grouped CatBoost ensembles using a 10-fold cross-validation committee¹⁸ to improve robustness and prediction diversity. Specifically, we partitioned the data so that all windows from a patient were assigned to a single fold, preventing patient leakage. The Base ensemble was trained on base (non-context) features, and the Full ensemble was trained on the complete feature set (base + temporal context features). For both, we trained a committee of 10 fold-specific models and repeated this procedure twice using different random permutations of the patient groups, yielding additional diversity across committee members. The High-sensitivity Full ensemble used the same full feature set but trained a single 10-fold committee; within each fold, Optuna¹⁹ was used to tune CatBoost hyperparameters (learning rate, depth, L2 regularization, bagging temperature, and random strength) to maximize validation sensitivity at a fixed threshold of 0.5.

Within each fold, we computed fold-specific class weights from the training data and passed these to CatBoost to mitigate label imbalance. Specifically, we used imbalance-aware training in CatBoost rather than resampling. Within each Train-fold, we computed class weights from the training portion only (inverse-frequency weighting of the seizure class with clamping to avoid extreme values) and passed these weights to CatBoost for all ensemble members. We did not perform over-sampling, under-sampling, or synthetic minority generation. The Development set was used only for selecting the operating point and event post-processing parameters, and the Eval set was used only once for final reporting. We did not use signal-level augmentation or synthetic EEG generation. At inference time, each committee member contributes a probability estimate; we first averaged probabilities within each ensemble, then computed the final window probability as an equal-weight average of the Base, Full, and High-sensitivity ensemble outputs.

Threshold selection and event post-processing

The detection pipeline operates as follows: (i) Window-level probabilities: The combined ensemble outputs a probability $p_i \in [0, 1]$ for each window i . (ii) Smoothing: Per-file, we apply a moving average filter to the sequence p_i with a window length corresponding to the smoothing duration (in seconds). This reduces short, isolated spikes in the probability sequence. (iii) Initial thresholding and dilation: The smoothed probabilities are thresholded at a decision threshold θ to create a binary mask of candidate seizure windows. Morphological dilation is applied in time to close small gaps between neighboring positive windows. (iv) Interval merging and pruning: Consecutive or nearby positive windows are merged into contiguous predicted seizure intervals, with a maximum allowed gap between windows. Predicted intervals shorter than a minimum duration are discarded.

All event-level decisions are made on top of 60-s windows with 75% overlap (15-s stride). For each recording, we apply a centered moving average over 3 consecutive window scores (kernel size = 3 windows at 15-s stride). The smoothed sequence is then thresholded at $\tau = 0.44$ to obtain a binary seizure mask. To reduce small gaps, we apply temporal dilation with a radius of two windows (± 30 s), and we subsequently merge candidate intervals separated by ≤ 40 s. Predicted intervals shorter than 10s are discarded. For event-level scoring, a ground-truth seizure is counted as detected if the corresponding predicted interval overlaps it by at least 10s.

To find the parameters for event detection, we conducted a joint parameter search on the development set. More specifically, on the Dev (development) set, we jointly search over: (i) Decision threshold τ , (ii) Smoothing window duration, (iii) Dilation radius, (iv) Maximum merge gap, (v) Minimum overlap with ground-truth events to count a detection, and (vi) Minimum predicted event duration. For each parameter combination, we compute event-level metrics on Dev. We enforced a minimum window-level specificity (specificity ≥ 0.93). An upper bound on false alarms per hour, based on a predefined value ($FA/h \leq 0.69$). Among configurations that satisfy these constraints, we selected the one that maximizes event sensitivity, with secondary tie-breaking based on time-based recall and positive predictive value. The selected parameters and threshold are then frozen and applied to the Eval split, where we compute the final window- and event-level metrics.

Rationale for alarm-budget constraints

Commercial seizure-detection software used in EMU video-EEG monitoring is typically characterized by an explicit detection-false-alarm trade-off, reported as false alarms per hour (FA/h). In a head-to-head evaluation of three commercial tools, Koren et al. reported mean FA/h values of roughly 0.2–0.9/h (overall mean ≈ 0.6 /h), with long-tail patients exceeding 6/h¹⁵. We therefore set an alarm-budget constraint of $FA/h \leq 0.69$ to align our operating point with the alarm burden observed in commercial EMU tools and enforced specificity ≥ 0.93 to avoid operating points dominated by false-positive windows that inflate alarm events after post-processing¹⁵.

Evaluation metrics and patient-level bootstrap CIs

We evaluated performance at two granularities: window-level (60-s windows, 15-s stride) and event-level.

Window-level (binary classification): Let TP, FP, TN, FN be counts of windows classified as seizure/non-seizure against the ground-truth window labels.

- Accuracy (ACC): $(TP + TN)/(TP + FP + TN + FN)$
- Sensitivity / Recall for seizure (Se): $TP/(TP + FN)$
- Specificity (Sp): $TN/(TN + FP)$
- Precision / PPV: $TP/(TP + FP)$
- Negative predictive value (NPV): $TN/(TN + FN)$
- Balanced accuracy (ACC_b): $(Se + Sp)/2$
- F1-score (F1): $2 \cdot (\text{Precision} \cdot \text{Recall})/(\text{Precision} + \text{Recall})$
- Matthews correlation coefficient (MCC):

$$\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

- AUROC: Area under the ROC curve (TPR vs. FPR) over all thresholds.
- AUPRC: Area under the precision-recall curve over all thresholds.
- Enrichment: PPV/ Prevalence.
 - where Prevalence is the percentage of seizures on Eval set (at the *window* level) = 7.17%.
- Macro vs. weighted averages (for Precision/Recall/F1): Macro = unweighted mean of per-class scores; Weighted = mean weighted by class support.

Event-level (interval detection): Post-processing converts window probabilities into predicted seizure intervals. A ground-truth seizure event is counted as detected if it overlaps at least one predicted interval by a minimum overlap duration δ .

- Event sensitivity (micro): $\frac{\text{\#detected ground-truth seizure events}}{\text{\#ground-truth seizure events}}$
- False alarms per hour (FA/h, micro): $\frac{\text{\#predicted events with zero overlap to any ground-truth seizure}}{\text{total recording hours}}$
- Time-based PPV (duration overlap precision): $\frac{\text{total overlap duration between predicted and true seizure intervals}}{\text{total predicted seizure duration}}$
- Time-based recall (duration overlap recall): $\frac{\text{total overlap duration between predicted and true seizure intervals}}{\text{total true seizure duration}}$
- Detection latency (reported descriptively): time from annotated seizure onset to the start of the first overlapping predicted interval; latencies are truncated at 0 s if the prediction begins before onset.

Confidence intervals: We compute 95% confidence intervals via patient-level bootstrapping: resample patients with replacement (including all their recordings/windows), recompute metrics per resample, and report the 2.5th – 97.5th percentiles over 1,000 bootstrap replicates.

Feature importance and selection of top features

To interpret the trained ensemble, we quantified feature importance using CatBoost's built-in importance scores. For each model in the cross-validated ensemble (all folds and repeats), we extracted per-feature importances based on how much each feature changes the model's predictions on average. We then aggregated these values by computing the mean importance for each feature across all ensemble members. Finally, features were ranked in descending order of this mean importance, and the top features are reported as the most influential inputs driving the detector's decisions (Figs. 4 and 5). Furthermore, we utilized SHAP values (Fig. 6) to determine the contribution and direction of base features (excluding temporal context features) to provide a more comprehensive explainability.

Data availability

The TUH EEG Seizure Corpus (TUSZ) 2.0.3 used in this study is publicly available from Temple University under its data use agreement. The dataset download/registration instructions and corpus landing page are available at: https://isip.piconepress.com/projects/nedc/html/tuh_eeg/index.shtml#c\tusz.

Code availability

The implementation of the model's inference pipeline will be shared publicly following publication. Until then, access can be provided by the corresponding author upon reasonable request, subject to institutional licensing constraints (<https://github.com/Mzabihi/A-Transparent-AI-Assurance-and-Benchmarking-Framework-for-EEG-Seizure-Detection-on-TUSZ>).

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Author contributions

MZ: Methodology, implementation, writing original draft. EG and KD: Methodology, writing - review & editing. ER: Methodology, supervision, writing - review & editing.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to M.Z.

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